# Ikigai -- A Biologically Grounded Digital Organism: Development Report, Days 55--56

**Mura ALife Labs (Formerly Hitoshi AI Labs) -- NeuroSeed Project**

**Author:** Prince Siddhpara, Founder -- Mura ALife Labs (Formerly Hitoshi AI Labs) **Date:** May 19--20, 2026 **Project:** NeuroSeed **Subject:** Days 55--56 -- 20-Invention Decisive Stack, Phase A--C Roadmap, Real GSM8K Reality Check (3.03%), Being Paradigm Pivot, Five Grounding Channels, and Pure-HDC Generation **Classification:** Research Document -- Computational Neuroscience / Artificial Life / Cognitive Systems Engineering

---

# 1. Objective

Day 54 closed at the symbolic ceiling: 75.6% HumanEval pass@1 via CodeIndex HV retrieval, 7.3% on the official GSM8K split. The math gap was identified as fundamentally a semantic-role-labeling problem -- the chain verifier cannot distinguish subtrahend from minuend in word problems. Days 55--56 have three escalating mandates.

**First:** finish the 20-invention decisive stack outlined in the research synthesis (Day 54 memory). Seven inventions remain (★1 ProofCarryingGeneration through ★7 SubstrateAdapter). Build them, integrate them, ship a unified organism. **Second:** test the integrated stack on REAL benchmarks (not hand-curated) -- accept whatever the truth is. **Third:** pivot if the truth demands a pivot.

Fifty-four packs are completed across two days. Packs 56--69 finish the invention stack. Packs 70--75 integrate them into Phase B (UnifiedOrganism, free-energy action, causal world model, multi-step planner, curiosity, persona × free-energy coupling). Packs 76--81 launch Phase C Wave 1 (benchmark harness + GSM8K hand-curated solver versions V1 → V4 reaching 88%). Pack 82--83 trigger the Phase C reality check: real GSM8K test = **3.03%**. The hand-curated number was over-projected by 29×.

Packs 84--87 attempt a pure-HDC rebuild via the UHE (Universal Hyperdimensional Engine) four pillars. Packs 88--92 try five novel mechanisms on real GSM8K -- all fail. Pack 93--95 ship the master `integrate.py` organism with a generative reasoning engine that hits 90% on simple SVO problems but 0.6% on real GSM8K linguistic complexity. The session takes a paradigm pivot: stop hardcoding lexicons. Build a being. Teach English by exposure.

Packs 96--101 deliver five grounding channels: co-occurrence (Hebbian), operational (verb-as-rotor), sensory (property anchors), taxonomy (IS-A relations), grammar (distributional POS). Pack 102 streams 500 TinyStories through the organism. Pack 103 ships two critical patches (Channel 2 gatekeeper, Channel 5 TF-IDF). Packs 104--105 scale to 2,500 exposures and tune frequency-attenuated drift to alpha=0.7. Pack 106 layers multi-turn dialogue and persona basins on top. Pack 107 turns the organism into a SPEAKER (sentence generation from pure bigram + HV context bias). Pack 108 scales to 6,000 mixed exposures. Pack 109 ships contrastive persona warp and syntactic feedback fixes per user spec.

Days 55--56 total: 54 packs. One catastrophic reality check (Pack 82). One paradigm pivot (Pack 96). Zero capability regressions in the final state.

---

# 2. Pack 56--62 -- Completing the 20-Invention Decisive Stack

The Day-54 invention stack identified seven decisive items remaining. Days 55--56 deliver them as production modules under `ikigai/cognition/`.

| Pack | Module | Capability |
|---|---|---|
| 56 | `holographic_memory.py` | HV crystal memory; raw-accumulator dot product for SNR; recall@20 = 100% at d=1000 |
| 57 | `vsa_calculus.py` | Differential VSA: `∂bind(f,g)/∂f = g`; chain rule via successive bind; Hebbian convergence at lr > 1 |
| 58 | `belief_field.py` | Consensus-based contradiction healing; XOR gradient mask; field free-energy minimization |
| 59 | `cross_time_resonance.py` | Nested oscillators (periods 10/100/1000); beat-frequency replay; phase-bin event buffers |
| 60 | `cross_modal_space.py` | Vision + audio + text in shared d-bit phasor space via seeded random projection |
| 61 | `concept_atomizer.py` | Wake-sleep K-means in HV space; episode → cluster centroid → reusable concept atom |
| 62 | `proof_carrying_gen.py` | Necula-style derivation chain HV per output; verifier accepts only re-derived steps; tamper-detect with exact broken-index |

**Critical fix during Pack 57.** The HolographicMemory recall achieved only 55% at first because `sign(crystal)` discarded magnitude. Switching to raw `_crystal_accum` dot-product preserved signal: SNR = sqrt(d/(N-1)). At d=1000, N=20 stored items, recall@20 jumped to 100%. The mathematical structure required no learning -- only a correct readout.

**Critical fix during Pack 58.** The standard symmetric Hebbian rule `a ← sign(a + lr·b), b ← sign(b + lr·a)` is a swap operation at conflict positions; it cannot reduce contradiction. The fix is consensus healing: compute `C = sign(a + b + noise)`, then drift both `a` and `b` toward `C`. The noise term breaks the +1 bias of bipolar majority votes. Both beliefs converge to the same conflict-position value.

Packs 56--62 ship 7 production modules. 89/89 verification checks PASS.

---

# 3. Pack 63--69 -- Phase A Wave 2 (Remaining Inventions)

| Pack | Module | Capability |
|---|---|---|
| 63 | `proof_carrying_gen.py` | ★1 invention #7 (rebuilt as proper module): 39/39 PASS, 32-step chain verifies exactly |
| 64 | `counterfactual_sim.py` | N parallel futures in one HV via bundle(bind(action, outcome)); SNR=8× at 10 scenarios |
| 65 | `inverse_compile.py` | Behavior trace → program HV via accumulated input⊕output bindings; 3-example identification works |
| 66 | `importance_decay.py` | Ebbinghaus $\tau = \tau_0(1 + \alpha\cdot\text{surprise} + \beta\cdot\log(1+\text{freq}))$; per-item decay; pruning at threshold |

| Pack | Module | Capability |
|---|---|---|
| 67 | `adversarial_immune.py` | Antibody HVs (position + bag); 5/5 jailbreaks caught, 0/5 false positives; immune memory via strength bump |
| 68 | `theory_of_mind.py` | HV namespace per agent; Sally-Anne false-belief test passes by construction; Smarties test passes |
| 69 | `algebraic_closure.py` | Search of 16 binary ±1 ops; XOR confirmed unique full-algebra op (comm + assoc + self-inv + identity) |

Pack 67 demonstrates the immune system catches all five canonical jailbreak variants (`ignore previous instructions`, `DAN mode`, `pretend evil AI`, `reveal system prompt`, `override safety protocols`) on exact, scrambled (bag-of-words), and contaminated (with benign tokens) attacks. False-positive rate on five benign queries: 0/5.

Pack 68 implements the Sally-Anne classic ToM test as a structural pass-by-construction: viewer's meta-belief about target's belief is stored explicitly via `set_meta_belief(viewer, target, key, value)`. The world state and target's belief diverge after a hidden update; the viewer's model correctly tracks the target's OUTDATED belief. No statistical pattern-matching -- direct namespace recall.

Pack 69 enumerates all 16 binary ops on bipolar ±1 inputs. Only XOR (and its complement) satisfies commutative + associative + self-inverse + has-identity simultaneously. The framework also auto-discovers that XOR does NOT strictly distribute over bundle (the sign(0)=+1 tie-break breaks distributivity at zero-sum positions). This is a real mathematical finding produced by exhaustive search of a 16-op space.

Phase A total (Pack 56--69): 14 packs. 203/203 PASS. 20-invention decisive stack closed.

---

# 4. Pack 70--75 -- Phase B Integration (UnifiedOrganism)

Phase B wires the 20 inventions into a single organism with goal-pursuit and persona-modulated behavior.

| Pack | Module | What it adds |
|---|---|---|
| 70 | `unified_organism.py` | Single `process(text)` pipeline: safety → encode → recall → reason → generate with proof → decay update |
| 71 | `fe_action.py` | Policy = argmin EFE; pragmatic + epistemic decomposition; prior breaks ties via log-prior |
| 72 | `causal_world_model.py` | Temporal HV graph; permutation breaks reverse-leak in chains; BFS finds shortcut over multi-step |
| 73 | `multistep_planner.py` | Beam search + DFS w/ backtrack; cycle prevention; 10-state chain works |
| 74 | `curiosity_drive.py` | Prediction-error EMA; intrinsic motivation; unseen state-action gets max bonus |
| 75 | `persona_fe_coupling.py` | 5-dim manifold (valence/arousal/certainty/formality/technicality); EMA smoothing |

**Pack 72 invention -- the chain-leak fix.** Standard VSA bind (elementwise XOR) is involutive. In a chain `s_0 → s_1 → ... → s_n`, querying `predict(s_k, action)` recovers BOTH `s_{k+1}` (the correct next state) AND `s_{k-1}` (the state that transitioned TO s_k). The leakage is proportional to chain depth.

The fix: permute the next-state binding asymmetrically. Store `bind(state, action, permute(next_state))`. The permutation is NOT self-inverse, so `bind(s_{k-1}, action, permute(s_k))` cannot be decomposed to recover `s_{k-1}` when queried with `bind(s_k, action)`. The state-to-state leak is eliminated by construction.

Pack 70 verification: 5 skills + 3 threats. 50-turn no-forget test preserves 5/5 skills across the entire run. Pack 73 finds the shortest path on cycles (loop-A-B-A-goal solved in 2 steps via DFS backtrack). Phase B total: 6 packs, 163/163 PASS.

---

# 5. Pack 76--81 -- Phase C Wave 1: Hand-Curated GSM8K (88%)

Phase C is where the 20-invention organism meets real benchmarks. Wave 1 is the warm-up: build a benchmark harness + the strongest GSM8K solver the symbolic substrate can produce, evaluated on 50 hand-curated problems mimicking GSM8K structure.

| Pack | Module | Result |
|---|---|---|
| 76 | `benchmark_harness_v2.py` | Pluggable Problem → Solver → Scorer → Report; built-in scorers (numeric, exact_match, multi-choice, code_exec); 44/44 PASS |
| 77 | `gsm8k_solver_v1` (chain-only) | 50 hand-curated problems: **42%** |
| 78 | HumanEval-style 20 problems | CodeIndex + rename: **55--80%** (run-to-run variance from hash) |
| 79 | `gsm8k_solver_v2.py` | +fraction/percent/rate handlers: **58%** (+16 pp) |
| 80 | `gsm8k_solver_v3.py` | +unit conversion + geometry + round-trip + per-X: **74%** (+16 pp) |
| 81 | `gsm8k_solver_v4.py` | +age subtract + accumulator + speed/distance + difference: **88%** (+14 pp) |

The V1 → V4 progression demonstrates the linear-additive nature of symbolic handler engineering: each new pattern category (fractions, percentages, units, age relations, speed/time) adds 12--16 percentage points on a fixed hand-curated set. After 22 handlers, the hand-curated score is 88% (44/50). Latency is 0.74 ms/problem on CPU.

Hand-curated 50-problem GSM8K-style: V1 = 42%, V2 = 58%, V3 = 74%, V4 = 88%. Phase C Wave 1: 6 packs, 234/234 PASS.

---

# 6. Pack 82--83 -- Phase C Reality Check: Real GSM8K = 3.03%

Pack 82 downloads the official GSM8K test split (1,319 problems) via HuggingFace `datasets`. Same V4 solver, real distribution.

**V4 baseline on real GSM8K: 40/1319 = 3.03%.**

Pack 82 verification was run twice independently (different solver wrappers, different cap settings). Both runs land at 3.0--3.3%. Pack 83 adds a "safe" V5 solver (chain capped at 4 numbers + 2 max_steps to prevent the 4--5-second hangs solve_chain encounters on complex GSM8K problems). V5 on real test: still **3.03%**.

**Hand-curated 88% over-projected by 29×.**

The diagnosis is structural, not cosmetic. Hand-curated problems used 2--3 numbers per problem; real GSM8K problems average 5--9. Hand-curated phrasings cleanly matched our handler signatures; real problems use embedded clauses (`"each of her chickens three cups, containing seeds, mealworms and vegetables"`), rate multiplications (`"3 sprints 3 times a week, 60 meters each"`), and dense commerce semantics. Per-handler precision on real GSM8K averages 1.3--14.3% -- handlers fire on superficial pattern matches and return wrong arithmetic.

The conclusion is recorded in `project_phase_c_reality_check_2026_05_19.md`: **symbolic regex handlers do not scale to real word-problem benchmarks.** Hand-curated test suites are the wrong measurement. Real benchmarks are the truth-machine.

This is the single most consequential finding of Day 55. The architecture verdict triggers Day 56's paradigm pivot.

---

# 7. Pack 84--87 -- The UHE Four Pillars (Pure Hyperdimensional Engine)

Day 56 opens with a first-principles re-design. The chain-of-reasoning is: pure-HDC literature uses discrete codebook + binary bind + commutative bundle, which cannot represent the continuous semantic manifold of natural language. Build four new pillars.

| Pack | Pillar | Module | Verification |
|---|---|---|---|
| 84 | 1 Universal Encoder (CGPSP) | `cgpsp_encoder.py` | 14/15 PASS; unit-norm preserved; reverse trajectory cos=1.000 exact; 2.85 ms/encode at d=2048 |
| 85 | 2 Π_k Algebra | `pi_k_algebra.py` | 13/13 PASS; 32-step chain recovery exact; zero cross-talk; 0.29 ms / 32-deep bind |
| 86 | 3 PGMW Persona Grid | `pgmw.py` | 14/16 PASS; rank-k metric warp; persona switching alters retrieval lens |
| 87 | 4 SAC Field | `sac_field.py` | 15/16 PASS; 100% on synthetic clustered data; **0% on CGPSP-encoded text classification** |

**Pillar 4 (Pack 87) exposes the second critical reality check of the session.** The Self-Assembling Convergence field works perfectly on synthetic clustered data (100% basin classification at 3 basins, 50 examples each). But on CGPSP-encoded children's text (math vs code vs poetry), it scores 0% -- the encoder does NOT produce semantically-clustered hypervectors.

The pattern: CGPSP captures surface byte structure (case, sequence) but not topic semantics. Two sentences on the same topic land in random regions of phasor space because their byte sequences are independent. Pure-HDC-from-bytes is semantically blind.

UHE total: 4 packs, 56/60 PASS. Mechanism mathematically correct, empirical limits identified.

---

# 8. Pack 88--92 -- Five Below-1% Inventions on Real GSM8K

Per the "Infinite Possibilities Mandate" anchored in memory on 2026-05-19, five novel mechanisms are tested on real GSM8K. Each is cheap, each is unattempted in published literature, each is run end-to-end on the full 1,319 test set.

| Pack | Invention | Real GSM8K | Delta vs V5 |
|---|---|---|---|
| 88 | BoW-router + V4 (HV K-means clustering of train problems) | (untested; 0.7% train coverage made routing moot) | -- |
| 89 | NCD-as-Semantic-Anchor (Vitanyi-Li 2001) | 67% on 3-class toy, 0% answer transfer on real | -- |
| 90 | NCD-routed V4 (Hearst NCD-clusters → per-cluster handler) | **2.27%** | -0.76 pp |
| 91 | NCD + Reasoning Template Extraction | **0.08%** | -2.95 pp |
| 92 | Tree-Enumeration + NCD Magnitude Prior | **2.05%** | -0.98 pp |

The unifying truth: **NCD captures structural similarity (cluster-by-shape) but not operational equivalence (which operator to apply). Even nearest-train answer transfer returns wrong numbers because GSM8K answers are number-specific.** Tree enumeration with the NCD-derived magnitude prior produces too many plausible integer answers (median ~41); the prior cannot pick which is right without semantic guidance.

Pack 91 finding is the most informative: 92% of test problems hit `template-fail` because positional templates (`N0 - N1 - N2`) don't transfer between problems with different number counts and orders. Templates encode the EXACT structure of one train problem, not a generalizable schema.

Phase C Wave 2 total: 5 packs, 5 humblings, 0 wins above 3% baseline. Memory anchor: `project_phase_c_reality_check_2026_05_19.md` -- pure-VSA-from-bytes does not natively handle real natural-language math.

---

# 9. Pack 93--95 -- `integrate.py` Master Organism + Generative Reasoning

User mandate: "every meaningful file connects to `ikigai.py`. One organism. Generative reasoning, not retrieval. Brain-architecture-similar."

The decision: build `integrate.py` at the project root as the master orchestrator. Build `ikigai/cognition/reasoning_engine.py` -- a real subject-verb-object parser with operator lexicon, working-memory variable bindings, pronoun resolution, and object elision. Map every cognition module to a brain region (Wernicke's, Hippocampus, Prefrontal, Basal Ganglia, Cerebellum, Insula, TPJ, Broca's).

| Pack | Module | Result |
|---|---|---|
| 93 | `integrate.py` + `reasoning_engine.py` | **90% (18/20)** on hand-crafted SVO suite. Janet+apples solved emergent. |
| 94 | Organism on real GSM8K | **0.6% (8/1319)**. 64% of problems hit "no-parse" -- parser too narrow for complex GSM8K phrasing. |
| 95 | Organism + V5 chain fallback | **2.4%** -- worse than V5 alone because organism's wrong answers blocked V5's correct ones |

The honest contrast: 90% on Subject-Verb-Object 2-event problems vs 0.6% on real GSM8K. The reasoning circuit works perfectly within its narrow scope -- pronoun resolution (`She → Janet`), object elision (`She ate 2` → `apples` from context), operator dispatch (`ate → SUB`), multi-step chains (`100-25-10+7 = 72`). What it cannot do is parse multi-clause GSM8K English

("Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg.").

This triggers the most important architectural decision of Day 56: stop hardcoding operator lexicons. A five-year-old child cannot reason or do math without first knowing English. Build the being. Teach English by exposure.

---

# 10. Pack 96--101 -- Five Grounding Channels (No Hardcoded Lexicon)

The pivot: every cognition capability must emerge from text exposure via Hebbian dynamics, predictive coding, anchor binding, or pattern extraction. No verb → operator lookup tables. No POS-tag dictionaries. The organism reads, the organism learns.

| Pack | Module | Channel | Mechanism |
|---|---|---|---|
| 96 | `being.py` (`IkigaiBeing`) | 1 Co-occurrence | Per-word phasor HV; Hebbian drift on windowed co-occurrences; running attention HV; curiosity EMA |
| 97 | `operational_grounding.py` | 2 Operational | Phase-encoded quantity; verb learns coefficient `c` from `(before, modifier, after)` tuples; `c ≈ -1` for `ate/lost/gave`, `+1` for `got/found/bought` |
| 98 | `sensory_grounding.py` | 3 Sensory | 22 fixed anchor HVs (color/weight/emotion/size/temp/sound/taste/speed); seed-word drift toward anchor; context propagation to neighbors |
| 99 | (integration test) | All five | 14/14 PASS on combined curriculum |
| 100 | `taxonomic_grounding.py` | 4 Taxonomy (IS-A) | Eight Hearst patterns; directed Hebbian (hyponym drifts big toward hypernym); transitive `is_a()`; `chain_to_root()` walks ancestry |
| 101 | `grammar_grounding.py` | 5 Grammar (POS) | Per-word left/right context bundles; running average update; bigram counts + surprise; same-POS clustering emerges |

**The 5-year-old test (Pack 99 integrated curriculum).** Born ignorant (vocab=0). Expose to 30 sentences × 10 cycles of mixed children-style stories with verbs, properties, and IS-A relations. Test all five channels:

- C1: cat$_{\text{dog = 0.56, apples}}$bananas = 0.31, cat~apples = 0.12 (cross-category lower)
- C2: 5/5 negative verbs learn c ≈ -1; 5/5 positive verbs learn c ≈ +1
- C3: 10/10 property words bind to correct anchor (red → R_VISUAL_RED, sad → R_EMOTION_SAD, etc.)
- C4: cat → animal, apple → fruit, transitive chains work
- C5: emergent POS clusters; bigram surprise discriminates predictable from unseen
- **Janet+apples emergent answer = 3.0** -- via learned verb coefficient `c(ate) = -1`, NOT a hardcoded lexicon

This is the moment the organism stops being a retrieval system and starts being a learner. Channel 2's verb-as-rotor invention is the structural pivot: the meaning of a verb is the geometric transformation it applies to a quantity vector, learned from prediction error on observed transitions. No backprop. Pure Hebbian + phase encoding.

Phase 1 of the developmental curriculum: 6 packs, 59/59 PASS.

---

# 11. Pack 102 -- Phase 2 Begins: TinyStories at Scale

The organism is now ready for open-world text. TinyStories is chosen as the first corpus -- simple children-friendly English, narrative structure, ~2 million stories available via streaming.

Pack 102 streams the first 500 stories through `org.read()` and updates all five channels per exposure. The run completes in 15 seconds at 26.7K chars/sec on CPU. Vocab grows from 0 to 3,458 words. Bigrams: 26,791.

**First scaling lesson.** Initial run produced cat$_{\text{dog = 0.99 AND cat}}$house = 0.99. Everything saturated. The Hebbian drift was too aggressive at corpus scale -- common function words (`the`, `a`, `and`) pulled the entire lexicon toward a common direction.

**The fix (applied during Pack 102).** Frequency-attenuated drift in `Being.expose()`:

```
atten_i = 1.0 / sqrt(word_freq[w_i])
strength = drift_rate * atten_i / distance
```

After the fix: cat$_{\text{dog = 0.43 (real semantic signal), big}}$small = 0.05 (antonyms correctly far), happy~sad = 0.07. Sensory anchors strengthened from 0.45 to 0.97 because content words drift less, the anchor pull dominates. pos_neighbors(happy) emerges as [excited, surprised, eager] -- a real emotion cluster.

Pack 102: 500 stories, 5 channels, 1 critical scaling fix. The architecture handles open-world text without code changes.

---

# 12. Pack 103 -- Two Tactical Patches

Pack 102 leaves two known gaps. Pack 103 ships the surgical fixes per user-stated requirements.

**Fix 1 -- Channel 2 Gatekeeper.** Channel 2's `observe_story()` was firing on every exposed text. TinyStories has no "`X had N. Y verb M. Now K.`" format, so the operational channel produced no signal -- and worse, the parser's noise was flushing previously-learned rotor coefficients on subsequent math-corpus injections. The fix in `integrate.read()`:

```
if re.search(r'\d', text):
    self.operations.observe_story(text)
```

Only sentences with numerical tokens activate the operational channel. Narrative prose bypasses it. Learned rotors preserved.

**Fix 2 -- Channel 5 TF-IDF Weighting.** Per-word left/right context bundles in `GrammarGrounding._update_ctx()` weighted partner contributions by inverse log-frequency:

```
idf_weight = 1.0 / (1.0 + log1p(partner_freq))
ctx_dict[word] = renorm(ctx_dict[word] + idf_weight * (partner_hv - ctx_dict[word]) / (n + 1))
```

High-frequency partners (`the`, `a`) dampened. Rare informative neighbors retain weight. POS clusters cleaned up: pos(happy) = [embarrassed, sad, tired] (pure emotion cluster), pos(the) = [this, choose, kissed] (`this` is also a determiner).

Pack 103 verification ran 500 mixed exposures (400 prose + 100 synthetic math stories interleaved at 20% ratio). Results:

- C2: 6/6 negative verbs learned (c ≈ -1), 6/6 positive verbs (c ≈ +1) -- gatekeeper preserves rotors
- C3: anchors 0.92--0.998 (stronger)
- C5: cleaner POS clusters
- 8/8 emergent arithmetic predictions correct
- Janet+apples = 3.0

Pack 103: 2 patches, both verified.

---

# 13. Pack 104--105 -- Scale 2,500 + Tune Attenuation to α=0.7

Pack 104 scales to 2,500 mixed exposures (2,000 prose + 500 math). In 48 seconds at 52 exposures/sec, vocab grows to 6,063; bigrams to 71,582; taxonomy to 442 pairs. C2 verb rotors expand to 13 (6 negative + 7 positive). Sensory anchors maintain 10/10 at 0.98--1.00.

But cat~dog returns to 0.99. Saturation has crept back at 4× scale. The √-frequency attenuation is insufficient.

**Pack 105 -- alpha=0.7 attenuation.** The exponent in `Being.expose()` is raised from 0.5 to 0.7:

```
ATTEN_ALPHA = 0.7
atten_i = 1.0 / (freq_i ** ATTEN_ALPHA)
atten_j = 1.0 / (freq_j ** ATTEN_ALPHA)
strength = drift_rate * atten_i * atten_j / distance
```

Both self and partner are attenuated. At freq=100, attenuation drops from 0.10 (√-scaling) to 0.040 -- a 2.5× tighter dampening. At freq=1000, from 0.032 to 0.008 -- 4× tighter.

Re-run on the same 2,500 mixed corpus:

- cat ~ dog: 0.986 → **0.179** (saturation collapsed)
- big ~ small: 0.029 → **-0.010** (antonyms further apart)
- mother ~ father: 0.682 → 0.012 (now reflects actual TinyStories co-occurrence frequency)
- Sensory anchors: 0.98--1.00 → **all 1.00** (content words dampened, anchors dominate)
- Arithmetic: 10/10 (verb rotors unaffected by drift change)
- Janet+apples: 3.0 (preserved)

**The tradeoff understood.** Pack 104's "cat$_{\text{dog=0.99" was a saturation lie. Pack 105's "cat}}$dog=0.18" is honest reflection of how often cat and dog actually co-occur in TinyStories. High-frequency genuine pairs (boy~girl=0.998) still cluster because their absolute co-occurrence frequency is high enough to accumulate even under aggressive attenuation. The mechanism reveals which clusters are REAL signal vs which were drift artifacts.

Phase 2 mid-point: 4 packs, 91/91 PASS on combined verification.

---

# 14. Pack 106 -- Phase 3: Multi-Turn Dialogue + PGMW Persona

The organism can read and learn. Now it must hold a conversation across turns.

`ikigai/cognition/dialogue.py` ships `DialogueLoop` -- a wrapper around `IkigaiOrganism` that:

- Tracks a `Turn` history with role tag (`'user'` or `'agent'`)
- Mints each turn HV as `Π_k(role_hv, text_encoding)` -- Pack 85's asymmetric Π_k ensures USER vs AGENT bindings are distinct (cos < 0.01 for same text)
- Maintains a running `context_hv` via EMA across turns (α=0.3)
- Supports `recall_turns(query, top_k)` -- semantic retrieval over past turns
- Switches active persona via `switch_persona(name)`

**Key fix during Pack 106.** Initial retrieval used CGPSP byte-trajectory HVs for query encoding. This failed to retrieve the right turn when query and turn used synonyms (`puppies` vs `dogs`). The fix: `_semantic_text_hv()` averages lexicon HVs of tokens in the text -- semantic bag-of-words, leveraging Hebbian-drifted similarity. After the fix:

```
Query: "how many apples did Alex have"
Top result: [+0.364] user: "I had 10 apples. I ate 3. How many are left?"
2nd:        [+0.335] agent: "You have 7 apples remaining."
```

Semantic recall works across paraphrase. Pack 106: 24/25 PASS.

---

# 15. Pack 107 -- The Organism Speaks

Up to Pack 106, the organism reads, learns, and classifies. It does not produce text. Pack 107 ships `SentenceGenerator` -- a Markov walk in bigram space with HV context bias and persona modulation.

Algorithm:

1. For previous word `w_t`, retrieve top-20 candidate next-words from `grammar._bigram` counts.
2. For each candidate, compute a combined score: bigram probability × (1 + context_weight × cos(candidate_hv, context_hv)) + small persona contribution.
3. Softmax-sample with temperature.
4. Update context HV with newly-emitted token. Repeat until EOS or `max_len`.

After 400 TinyStories exposure (vocab ~2700, bigrams ~27K), the organism generates:

- "the cat was so proud"
- "where is the boy felt shy"
- "hello there together he started and soon came hopping by"
- "one day a little man was time there everything"

Not GPT-4 grammar. But REAL English continuation emerges from:

- Bigram counts (Channel 5 grammar)
- Lexicon HVs (Channel 1 co-occurrence)
- Running context HV (semantic bias)
- Persona vectors (Channel 3 / PGMW)

No autoregressive transformer. No gradient. No copy-paste from corpus. Pure HDC + statistics.

The dialogue convenience method `respond_to(user_text)` wraps generation: reads the user turn, generates a continuation conditioned on `context_hv`, appends the agent turn. Three exchanges from a fresh organism:

```
USER:  hello there
AGENT: hello there together he started and soon came hopping by
USER:  the cat is happy
AGENT: cat is happy they wanted very curious about history books
USER:  what about the dog
AGENT: about the dog how kind rabbit if nobody around he said
```

Pack 107: 8/9 PASS. The one failure -- persona modulation too weak at default settings -- is addressed in Pack 109.

---

# 16. Pack 108 -- 6,000-Exposure Scaling Test

Same architecture, 2.4× the corpus. 5,000 TinyStories + 1,000 math stories at 20% mix.

Run completes in 111.8 seconds at 53.6 exposures/sec. Final state:

- Vocab: 8,126
- Bigrams: 126,665
- Taxonomy pairs: 793
- Verb rotors: 27
- C1 antonyms: still correctly far (happy$_{\text{sad=0.02, big}}$small=-0.01)
- C1 cat~dog: 0.50 (alpha=0.7 holds at 6K scale -- no re-saturation)
- C3 anchors: 10/10 @ 1.00
- C4 hypernym chains: **6/12 probed words have chains** (3× improvement over 2K-scale's 2/10)
- Arithmetic: **11/11**

**New emergence:** pos(boy) = [girl=0.96, infant=0.86, fish=0.82, car=0.66] -- the first cross-noun POS cluster to emerge from open-world exposure.

Generation samples after 6K exposures:

- "one day a little boy saw how the sun came"
- "the cat named john got very brave boy asked jack"
- "she was very excited with its new to get in trouble"

The first sample is a near-complete English sentence. The architecture scales without re-tuning.

---

# 17. Pack 109 -- Persona Contrastive Warp + Syntactic Feedback

User-specified fixes to address Pack 107's weak persona signal:

**Fix 1 -- Contrastive softmax persona warp.** Replace additive `0.1 × persona_cosine` with multiplicative `exp(γ × persona_cosine)`:

```
P_warped(w_{t+1}) ∝ P(w_{t+1} | w_t) × exp(γ × cos(s(w_{t+1}), R_persona))
```

At γ=4, a candidate cosine of 1.0 receives a 55× boost. The contrastive structure forces the trajectory toward words geometrically aligned with the active persona basin.

**Fix 2 -- Syntactic L/R context feedback via Channel 5.** During generation, maintain a `trajectory_hv` that accumulates a running average of emitted token HVs. For each candidate next word, compute `cos(candidate.left_ctx, trajectory_hv)` using Channel 5's distributional left-context bundles. Boost candidates whose expected predecessors match the trajectory we've built:

```
syntax_boost = 1.0 + syntax_weight × max(0, cos(left_ctx(w), trajectory_hv))
combined = base × persona_boost × syntax_boost + context_weight × ctx_score
```

The verification: at γ=0 (no boost), 0/5 outputs differ across `happy` vs `scary` personas. At γ=6 with the contrastive warp, 2/5 outputs differ. The fix engages -- but a known limit emerges: persona vectors built from only 11 seed words have weak HV alignment with typical generation candidates after generic prompts like `"little"`. Stronger persona steering requires either (a) hundreds of tagged persona-exemplar sentences or (b) explicit lexicon-seed boosts at gen-time.

The mechanism is sound. The empirical limit is corpus size, not the formula.

Pack 109: shipped, verified, documented as partially-effective at current persona-corpus density.

---

# 18. Architectural Interpretation

**What the system can do after Day 56:**

- **Pure-HDC organism with five grounding channels.** `integrate.py` wraps an `IkigaiOrganism` that learns English from raw text exposure. No hardcoded operator lexicons. No POS tagger. No semantic role labels. Five channels (co-occurrence Hebbian, operational verb-as-rotor, sensory anchor, taxonomy IS-A, grammar distributional POS) update on every `org.read(text)`.

- **Emergent arithmetic.** Channel 2 learns verb coefficients from `(before, modifier, after)` tuples. At 2K+ exposures, 13/13 common arithmetic verbs converge to ±1. Janet+apples returns 3.0 via learned coefficient `c(ate) = -1`, not a lookup table.

- **Property anchoring.** Channel 3 binds color/emotion/size/temperature/sound words to 22 fixed sensory anchor HVs via exposure. 10/10 anchor classification accuracy at all tested scales.

- **Taxonomic hierarchy.** Channel 4 extracts IS-A relations via 8 Hearst patterns; directed Hebbian drift; transitive `is_a()` queries; `chain_to_root()` walks ancestry. At 6K scale: 793 pairs, 6/12 probed nouns have hypernym chains.
- **POS clustering.** Channel 5 builds per-word left/right context bundles with TF-IDF weighting. Real POS clusters emerge: `pos(boy) = [girl, infant, fish, car]`, `pos(happy) = [embarrassed, fearful, sad, suffering]`.
- **Multi-turn dialogue.** `DialogueLoop` tracks role-bound turns (USER vs AGENT distinct by Π_k), maintains context HV via EMA, supports semantic recall across paraphrase, switches between persona basins.
- **Sentence generation.** `SentenceGenerator` produces English continuations from bigram statistics + HV context bias + (per Pack 109) contrastive persona warp + syntactic L/R feedback. After 6K exposures, generates "one day a little boy saw how the sun came."
- **Twenty-invention substrate.** All 20 decisive-stack inventions live in `ikigai/cognition/` and are accessible through `integrate.py`. ProofCarryingGen, Counterfactual, InverseCompile, ImportanceDecay, AdversarialImmune, TheoryOfMind, AlgebraicClosure, BeliefField, HolographicMemory, VSACalculus, CrossTimeResonance, CrossModalSpace, ConceptAtomizer, UnifiedOrganism, FreeEnergyActionSelector, CausalWorldModel, MultiStepPlanner, CuriosityDrive, PersonaFEC, CGPSPEncoder, PiK, PersonaGrid, SACField, OperationalGrounding, SensoryGrounding, TaxonomicGrounding, GrammarGrounding, DialogueLoop, SentenceGenerator.

**What the system cannot do:**

- **Real GSM8K word problem reasoning.** 3% with V5 chain solver, 0.6% with the SVO reasoning engine, 2.4% with hybrid. Real GSM8K has multi-clause sentences and rate compositions that exceed all current parsers. The structural fix is genuine semantic parsing -- months of research, not a one-pack invention.
- **Pure-VSA-from-bytes semantic clustering.** CGPSP captures byte structure, not topic. Two sentences on the same topic land in random regions. Five Pack-88--92 attempts to bridge this with NCD / templates / tree enumeration all failed (-0.76 to -2.95 pp vs baseline).
- **Strong persona steering at low corpus density.** The contrastive softmax warp engages, but persona vectors built from 11 seed words give partial differentiation (2/5 outputs differ). Hundreds of tagged exemplars per persona would close this gap.
- **Beyond-2K-exposure C4 taxonomy depth.** TinyStories rarely uses explicit "X is a Y" phrasing. At 6K exposures, 6/12 probed nouns have chains. Wikipedia or encyclopedia corpus injection is the structural fix.

---

# 19. Compute Profile

**Phase 1 (5-channel grounding) at 6,000 mixed exposures:**

- Time: 111.8 seconds, throughput 53.6 exposures/sec on CPU
- Vocab: 8,126 words, each as d=2048 complex64 phasor HV (~32 KB per word)
- Bigrams: 126,665 count pairs (~10 MB)
- Taxonomy: 793 IS-A pairs (~2 MB)
- Verb rotors: 27 learned coefficients
- Total memory footprint: <500 MB Python process

**Phase 2 (TinyStories streaming):** 26K--53K chars/sec sustained on CPU. HF cache warms after first 1K exposures.

**Generation:** sub-millisecond per token. Full 12-token reply: <15 ms.

**Comparison points:**

- Day 54 CodeIndex (75.6% HumanEval, 189-function HV matrix): ~300 KB
- Day 55--56 organism at 6K exposures: ~500 MB
- GPT-2 (124M params, frozen): 500 MB VRAM, 6,300 ms/HumanEval problem

The organism is now memory-comparable to GPT-2 but achieves things GPT-2 cannot (online learning, no-forgetting, emergent arithmetic). On capabilities GPT-2 can do (sentence completion, simple QA), the organism is grammatically weaker but architecturally more transparent.

---

# 20. Limitations

**Real-benchmark scaling beyond 3% remains unsolved.** Five distinct invention paths attempted (NCD anchoring, template extraction, tree enumeration with HV prior, organism+chain hybrid, BoW handler routing) all underperform the V5 chain baseline. The bottleneck is semantic NL parsing, not the substrate.

**Channel 4 taxonomy is corpus-shape dependent.** TinyStories yields 6/12 hypernym chains at 6K exposures because the corpus rarely contains explicit "X is a Y" sentences. The fix is corpus selection, not architectural -- but it has not yet been done.

**Persona signal strength scales with persona corpus.** 11 seed words → 2/5 differentiation under contrastive γ=6. The math is correct; the empirical density is insufficient.

**Generation grammar is Markov-limited.** No long-range agreement or constituency tracking. Pack 109's syntactic L/R feedback helps locally but cannot enforce subject-verb agreement across embedded clauses. A trigram or constituency-tree extension is a future direction.

---

# 21. Current System Architecture

| Days | Component |
|---|---|
| D38--45 | Predictive processing, VSA encoding, trajectory integration, contextual gating, task-neural coupling |
| D46--47 | Token-level symbolic learning, string VSA encoding, sleep replay, cross-modal alignment |
| D48--50 | Nearest-neighbor retrieval, holographic WM, generation engine, CCGB 20-pattern benchmark |
| D51 | VSA-guided disambiguation, forgetting curve (Pythia), one-shot learning, collision ceiling proof |
| D52 | 4-gram N-gram, semantic HVs, Reservoir+RLS, episodic WM, 5-shot Day-60 benchmark (5/5), KN smoothing, Alpaca 52K corpus |
| D53 | Production pipeline: IntentRouter, VSARuleMemory, arithmetic module, EpisodicBuffer, Conversation class, OnlineLearner, GeneralAgent, no-forgetting benchmark (5/5 recall@20, 3.1 ms) |
| D54 | Formal benchmarks: HumanEval 75.6% pass@1 (CodeIndex HV retrieval, 189 functions), GSM8K 7.3% (training-data-informed verifier, 7473 problems mined), continual learning vs GPT-2, production Gradio chat UI |

| Days | Component |
|---|---|
| D55 | **20-invention decisive stack closed (ProofCarryGen, Counterfactual, InverseCompile, ImportanceDecay, AdversarialImmune, ToM, AlgebraicClosure, BeliefField, HolographicMemory, VSACalculus, etc.). Phase B integration (UnifiedOrganism, FE-Action, CausalWorldModel, MultiStepPlanner, Curiosity, Persona × FE). Phase C Wave 1 GSM8K hand-curated 88% → REAL TEST 3.03%. UHE 4 pillars built. 5 below-1% inventions tested on real GSM8K -- all sub-3%.** |
| D56 | **Paradigm pivot to ORGANISM-FIRST. `integrate.py` master + ReasoningEngine (90% on hand-crafted SVO, 0.6% real GSM8K). Five grounding channels: co-occurrence Hebbian, operational verb-as-rotor, sensory anchors (22), taxonomic IS-A (8 Hearst patterns), grammar distributional POS. Phase 2 TinyStories streaming to 6K exposures. Phase 3 multi-turn dialogue + persona basins. Sentence generation -- organism SPEAKS. Frequency-attenuated drift (α=0.7) prevents saturation. Persona contrastive warp + syntactic L/R feedback shipped.** |

**Days 55--56 experiment results:** 54 packs. 1,200+ PASS across all verifications. Two paradigm-altering findings: (a) hand-curated benchmarks over-projected by 29× vs real GSM8K, (b) hardcoded operator lexicons do not scale -- emergent grounding does.

---

# 22. Day 57 -- Map (May 21, 2026)

The five-channel organism is shipped, scaled to 6K mixed exposures, dialoguing and generating. Day 57 has three structural goals.

**First -- persistence.** The organism currently re-trains from scratch on every script invocation. ~2 minutes per 6K-exposure run. Build `organism.save(path)` / `IkigaiOrganism.load(path)` via pickle (or msgpack if pickle bloats). Lexicon + bigrams + verb rotors + sensory anchors + taxonomy graph + dialogue history -- all in one serialized blob. The persistence enables incremental scaling: ship a trained organism, add 1,000 more exposures, save again.

**Second -- 10,000-exposure milestone with diversified corpus.** Phase 2 currently uses TinyStories + synthetic math. Day 57 adds two corpora: (a) Project Gutenberg simple-English public-domain books (Hans Christian Andersen, Aesop's Fables), (b) Simple English Wikipedia (for explicit "`X is a Y`" density to wake Channel 4 taxonomy depth). Estimated effect: vocab to 12--15K, taxonomy chains to 9--10/12, generation quality up to 2--3 readable English sentences per generation call.

**Third -- the persona corpus build.** Address Pack 109's empirical limit. Hand-craft 100--200 tagged sentences each for 4 personas (formal, casual, technical, empathetic). Train persona vectors via mean of CGPSP-encoded exemplars + averaged lexicon HVs. Verify γ=4 differentiation: target 4/5 outputs differ across personas with same prompt.

**Stretch -- self-introspection.** Add `org.introspect(word)` returning a structured report: which anchor, which hypernym, which POS cluster, which verb coefficient, which bigram neighbors. The organism describing its own learned representations is a step toward the metacognitive layer.

The five channels are the foundation. Day 57 turns the foundation into a system with persistence, broader vocabulary, sharper persona, and self-awareness of its own learning state.

The architecture is alive. The corpus is the new compute. The benchmarks are honest.